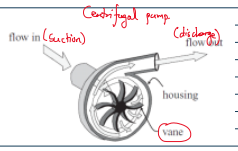
Pump Characteristics Curves

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.8)

- Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement
  - Working principle is energy balance! Electric energy → shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics
  - Pump manufacturers need to characterise a pump → Pump operation under various conditions [Pump curves!]



$\Delta E = Q + W$ (total energy of system; Heat from environment to the system via boundary; Work done on system by environment)

Energy balance on CV: $\frac{dE_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\vec{v})\rho\hat{E}\,dS = \dot{Q}_{net} - \dot{W}_{net,CV}$ $\quad [E_{CV} = U + E_K + E_p]$

Closed system

Open system (mass flow in & out)

Some assumptions

$\Delta E_p + \Delta E_K + \Delta U = \dot{Q}_{in,CV} - \dot{W}_{by,CV}$ $\quad (\dot{W}_f = \vec{F}\cdot\vec{v} = (\vec{P}\cdot\vec{A})\cdot\vec{v})$

$\dot{W}_s$ Shaft work (work performed by fluid due to moving parts)

$\dot{W}_f$ Flow work (work done by fluid as it enters & leaves the system)

$H = U + PV$

SS Main Energy balance:

$$\Delta E_p + \Delta E_K + \Delta H = \dot{Q}_{in,CV} - \dot{W}_{s,by\,CV}$$

Single in, single out, incompressible fluid

$$\dot{m}\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + \dot{m}g(z_2 - z_1) + \dot{m}\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,CV} - \dot{W}_{s,by\,CV}$$

Mechanical energy Balance (MEB)

Temp constant, no phase changes/chemical reaction, negligible heat transfer

pressure

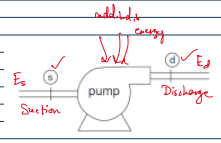
$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{\dot{W}_{s,by}}{\dot{m}}$$

ρ density; $\langle v\rangle$ average velocity; $\alpha = 1$ (turbulent flow); F friction term; $[\Delta U = Q]$

laminar

turbulent

Pump can add energy via:
- Increase velocity of fluid
- create pressure differential
- Pump can change fluid height (raise its potential energy)



η = Pump efficiency = $\frac{\text{(useful energy)}}{\text{(total energy)}}$ (energy converted to work (rather than friction))

$$= \frac{\left(-\frac{\dot{W}_{s,by}}{\dot{m}} - F\right)}{\left(-\frac{\dot{W}_{s,by}}{\dot{m}}\right)}$$

Pump brake horsepower: $bhp = -\dot{W}_{s,by}$ (Total work delivered by pump)

bhp & $\dot{m}$

Dividing by g

$$\therefore \frac{\Delta P}{\rho g} + \frac{\Delta\langle v\rangle^2}{2\alpha g} + \Delta z = \eta\left(-\frac{\dot{W}_{s,by}}{\dot{m}}\right) = \eta\,(bhp)$$

Δ = out − in

outlet

inlet

$$\frac{\Delta p}{\rho g} + \frac{\Delta\langle v\rangle^2}{2\alpha g} + \Delta z = \eta\left(\frac{-W_{s,by}}{m}\right) = \eta\,(bhp)$$

Δ = out − in

$$\left(\frac{p_d}{\rho g} + \frac{\langle v\rangle_d^2}{2g} + z_d\right) - \left(\frac{p_s}{\rho g} + \frac{\langle v\rangle_s^2}{2g} + z_s\right) = \eta\left(\frac{-W_{s,by}}{mg}\right)$$

outlet — inlet

Discharge head (H_d) [=] length — Suction head (H_s) [=] length

$$(H_d - H_s) = \frac{L}{\eta}\left(\frac{-W_{s,by}}{mg}\right) = \eta\left(\frac{-bhp}{mg}\right)$$

Head loss

give info about $\dot{m}$ & bhp

Head loss (ft) — @ fixed RPM — Q (Capacity) — related to $\langle v\rangle$?

Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
+ Shape of vanes
+ flow path

Lab: Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP
↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables & then plot them!

ΔH = Head loss; $\frac{\Delta p}{\rho g}$; Δz; $\frac{\Delta\langle v\rangle^2}{2g}$; Q

discharge; PS; Pump; Suction

PROBLEM: 8.30 Calculate the extra stress tensor $\underline{\underline{\tau}}$ for potential flow around a sphere of an inviscid, Newtonian fluid. Calculate the total stress tensor $\underline{\underline{\Pi}}$. Comment on what is obtained.

inviscid → $\mu = 0$ → Shear stress = 0

$$\underline{\underline{\tau}} = 0$$

$$\rho\left(\frac{\partial \underline{v}}{\partial t} + \nabla\underline{v}\cdot\underline{v}\right) = \mu\nabla^2\underline{v} + \rho\underline{g} + \nabla p$$

$\mu\nabla^2\underline{v} = 0$

$$\underline{\underline{\Pi}} = -p\underline{\underline{\delta}} + \underline{\underline{\tau}}^{\,0}$$

dynamic pressure $\hat{p}$

Addison Cycle

Fluid 1 (μ_1)

Interface

Fluid 2 (μ_2) → $\tau_1 = \tau_2$

Air ($\mu = 0$)

Fluid 1 (μ_1) → τ @ interface = 0

PROBLEM: 8.40 What is the force (i.e., drag) on a thin plate given the following conditions? The fluid is water, the plate is 0.52 m long and 6.3 m wide, and the free-stream velocity is 1.3×10^{-2} m/s

$v_{inlet} = v_\infty$ (μ)

$F_r = \iint (\underline{n}\cdot\underline{\underline{\pi}})\big|_w\, dS$

6.3 m

0.52 m

$v \propto x_1$

Boundary layer flow:

$\nabla\cdot\underline{v} = 0$

Navier Stokes

v varying w/ two dimensions

Assumptions

PDE

Velocity profile for laminar flow past a flat plate

non-dime x_1

$$\frac{v_1}{v_\infty} = f'(\zeta) = 1 - (0.5434)\log\left[1 + (68.3)10^{-0.6247\zeta}\right]$$

Navier Stokes ✓

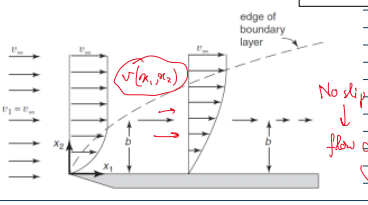
dimensions

laminar flow past a flat plate

$$\vec{F} = \iint_{Surface} [\hat{n} \cdot \pi] \, dS$$

$$\tau_w = 0.332 \mu v_\infty \left(\frac{\rho v_\infty}{\mu x} \right)^{1/2}$$

Drag is the force component in flow direction



No slip → flow changes → v

$$\left(\frac{\tau_w}{0.332} \right)^2 = \mu v_\infty^2 \left(\frac{\rho v_\infty}{\mu x_1} \right)$$

$$[\tau] = \mu v_\infty^3 \rho \, x_1$$

$$\frac{kg}{m s} \cdot \frac{m^3}{s^3} \cdot \frac{kg}{m^3} \cdot m = \frac{kg^2}{m^2 s^4}$$

$$\tau^2 \quad [\tau] \quad p_a^2$$

$$\frac{kg^2}{m^2 s^4}$$

$$\sum F = m \cdot a \quad [F] \quad \frac{kg \cdot m}{s^2}$$

$$p_a = \frac{F}{A} = \frac{N}{m^2} = \frac{kg \cdot m}{m^2 s^2} = \frac{kg}{m s^2}$$

$$Stress = \frac{F}{A}$$

$$F = \tau_w \cdot A$$